AI-Powered Productivity Dashboard

Final Project Report

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Abstract

This report presents a full■stack AI■powered Productivity Dashboard that combines task, reminder, and meeting management with a locally hosted AI assistant via Ollama. The system offers CRUD operations, authentication, and a responsive modern UI. It is designed for extensibility and future enhancements such as streaming responses and OAuth integration.

Introduction

The AI Productivity Dashboard bridges traditional task tools with intelligent assistance. Users can maintain their daily workflow while leveraging contextual AI answers like “What’s due today?” or “List my meetings for tomorrow,” powered by structured data from the application.

Objectives

- Provide CRUD functionality for tasks, reminders, and meetings.
- Integrate a local AI assistant (Ollama) for contextual help.
- Implement authentication with protected routes (JWT/Custom).
- Deliver a responsive, dark■mode UI with clean UX.
- Enable straightforward deployment (Vercel FE + Ngrok/VPS BE).

System Architecture

The system consists of a React frontend (Vercel), a Node.js/Express backend (Ngrok/VPS), and a locally hosted AI layer (Ollama). The frontend communicates with the backend over REST APIs; the backend proxies chat requests to Ollama, enriching prompts with user context (tasks, reminders, meetings).

High-Level Flow:

- [User] → [Frontend – React] → [Backend – Node/Express] → [Ollama API]
- Backend aggregates user context and crafts the system/user prompts for Ollama.
- Responses return to the chat UI and can reference live task/reminder data.

Technology Stack

Layer	Technology
Frontend	React + Vite + Tailwind CSS
Backend	Node.js + Express
AI Model	Ollama (LLaMA3, Mistral, Gemma)
Auth	JWT / OAuth / Custom
Deployment	Frontend: Vercel | Backend: Ngrok / VPS
Data Storage	In■memory / JSON / Database (future)

System Modules

- User Interface (React + Tailwind): Sidebar navigation, task/reminder/meeting panels, chat modal, dark mode.
- Backend API (Node.js): REST endpoints for auth, tasks, reminders, meetings, and chat.
- AI Assistant (Ollama): Context■aware responses using system+user prompts.
- Authentication: Login/Signup with JWT & protected routes.
- Deployment: Vercel for FE, Ngrok/VPS for BE + Ollama.

AI Integration

Ollama hosts local LLMs (e.g., LLaMA3, Mistral, Gemma). Each chat request includes a system prompt describing the assistant's role and a user prompt with the message. When available, recent tasks/reminders are embedded as context to ground responses. The backend can switch models dynamically.

Features

- CRUD for tasks, reminders, meetings.
- AI chat assistant with contextual awareness.
- Modern, responsive UI with dark mode.
- Protected routes and token■based auth.
- Configurable API base URL for deployments.

Future Enhancements

- Stream AI responses (Server■Sent Events / WebSockets).
- Persist chat history (localStorage or DB).
- Voice input (Web Speech API) and TTS.
- Live context sync as user edits tasks.
- Google OAuth and enterprise SSO.

Conclusion

The AI Productivity Dashboard blends robust task management with local, privacy■friendly AI assistance. Its modular architecture supports incremental evolution toward real■time streaming, richer storage, and wider integrations.

Appendix – Folder Structure

/client (React)

/components

/pages

/assets

App.jsx

index.jsx

/server (Node.js)

server.js

/routes

chat.js

tasks.js

auth.js

/data

userData.json